Propionamide $C_2H_5-\underset{\underset{O}{\|}}{C}-NH_2$ Propanamide	–	prop	ane	amide
Benzamide $C_6H_5-\underset{\underset{O}{\|}}{C}-NH_2$ Benzamide	–	benz	_	amide

12. 14. 2. Acid Halides:

Methods of Preparation of acid chloride:

Acid chlorides are prepared from carboxylic acid by treating it with anyone of the chlorinating agent such as $SOCl_2$, PCl_5, or PCl_3

1) By reaction with thionyl Chloride ($SOCl_2$)

$$\underset{\text{Acetic acid}}{CH_3-\overset{\overset{O}{\|}}{C}-OH} + SOCl_2 \longrightarrow \underset{\text{Acetyl chloride}}{CH_3-\overset{\overset{O}{\|}}{C}-Cl} + HCl + SO_2$$

This method is superior to others as the by products being gases escape leaving the acid chloride in the pure state.

Physical properties:

- They emit pale fumes of hydrogen chloride when exposed to air on account of their reaction with water vapour.
- They are insoluble in water but slowly begins to dissolve due to hydrolysis.

Chemical properties:

They react with weak nucleophiles such as water, alcohols, ammonia and amines to produce the corresponding acid, ester, amide or substituted amides.

1) Hydrolysis. Acyl halides undergo hydrolysis to form corresponding carboxylic acids

$$\underset{\text{Acetyl chloride}}{CH_3-\overset{\overset{O}{\|}}{C}-\boxed{Cl + H}OH} \longrightarrow \underset{\text{Acetic acid}}{CH_3-\overset{\overset{O}{\|}}{C}-OH} + HCl$$

XII U12-Carbonyl Compounds.indd 182 11-01-2021 16:10:47